Indexes work best when they are highly selective, meaning they select a small proportion of the total number of rows. If only 10% of the rows have a value of true, then a query that filters on that boolean column for True values will likely produce an Index Scan and perform well.

Now you're ready to work with some of the less common Index types. First up is the Expression Index.

Transform Values with an Expression Index

An Index that covers a column may transform the column value before being stored. When this type of transformation occurs the Index is called an Index on an Expression. This may also be called a Functional Index because a database function is being called, like LOWER().

Imagine that want to search users by email address in a case insensitive way. One solution would be to transform the email address using the LOWER() function on the column before it's stored. Use the temp.schema in Rideshare that you set up earlier for the example you'll find next. If you need to recreate this schema and table, refer to the code example called "insert_users_generate_series.sql" from an earlier chapter. When you've done this, you'll have the temp schema and temp.users table with 10 million rows populated.

Run this SQL statement from psql to insert more rows into temp.users:

```
INSERT INTO temp.users (
first_name, last_name, email, type, created_at, updated_at
)
VALUES
  ('Jane', 'Example1', 'Jane@example.com', 'Driver', NOW(), NOW()),
  ('jane', 'example2', 'jane@example.com', 'Driver', NOW(), NOW());
```

Email addresses are case insensitive, so you've created duplicate email addresses now in your system. You want know which Jane should get an email, whether it should be jane% or 'Jane%'. This creates a problem and you can use a Unique Index on an Expression to solve this.

First, remove the users you just created by running this statement:

```
DELETE FROM temp.users
WHERE email IN ('Jane@example.com', 'jane@example.com');
```

After improving your data model by validating that you only accept unique case insensitive email addresses, you'll try and add the users back.